In [ ]:

```
import pandas as pd
dataset=pd.read_csv("Startup_dataset.csv")
dataset
```

In [10]:

```
dataset=dataset.drop(columns=["Unnamed: 0"])
dataset
```

Out[10]:

	RD Spend	Administration	Market Spend	State	Profit
0	165349.30	136897.90	471784.20	New York	192261.93
1	162597.80	151377.69	443898.63	California	191792.16
2	153441.61	101145.65	407934.64	Florida	191050.49
3	144372.51	118671.95	383199.72	New York	182902.09
4	142107.44	91391.87	366168.52	Florida	166188.04
5	131877.00	99814.81	362861.46	New York	156991.22
6	134615.56	147198.97	127716.92	California	156122.61
7	130298.23	145530.16	323876.78	Florida	155752.70
8	120542.62	148719.05	311613.39	New York	152211.87
9	123334.98	108679.27	304981.72	California	149760.06
10	101913.18	110594.21	229161.05	Florida	146122.05
11	100672.06	91790.71	249744.65	California	144259.50
12	93863.85	127320.48	249839.54	Florida	141585.62
13	91992.49	135495.17	252665.03	California	134307.45
14	119943.34	156547.52	256513.02	Florida	132602.75
15	114523.71	122616.94	261776.33	New York	129917.14
16	78013.21	121597.65	264346.16	California	126993.03
17	94657.26	145077.68	282574.41	New York	125370.47
18	91749.26	114175.89	294919.67	Florida	124267.00
19	86419.80	153514.21	0.10	New York	122776.96
20	76253.96	113867.40	298664.57	California	118474.13
21	78389.57	153773.53	299737.39	New York	111313.12
22	73994.66	122782.85	303319.36	Florida	110352.35
23	67532.63	105751.13	304768.83	Florida	108734.09
24	77044.11	99281.44	140574.91	New York	108552.14
25	64664.81	139553.26	137962.72	California	107404.44
26	75328.97	144136.08	134050.17	Florida	105733.64
27	72107.70	127864.65	353183.91	New York	105008.41

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	RD Spend	Administration	Market Spend	State	Profit
28	66051.62	182645.66	118148.30	Florida	103282.48
29	65605.58	153032.16	107138.48	New York	101004.74
30	61994.58	115641.38	91131.34	Florida	99937.69
31	61136.48	152702.02	88218.33	New York	97483.66
32	63408.96	129219.71	46085.35	California	97427.94
33	55494.05	103057.59	214634.91	Florida	96779.02
34	46426.17	157694.02	210797.77	California	96712.90
35	46014.12	85047.54	205517.74	New York	96479.61
36	28663.86	127056.31	201126.92	Florida	90708.29
37	44070.05	51283.24	197029.52	California	89949.24
38	20229.69	65948.03	185265.20	New York	81229.16
39	38558.61	82982.19	174999.40	California	81005.86
40	28754.43	118546.15	172795.77	California	78240.01
41	27893.02	84710.87	164470.81	Florida	77798.93
42	23641.03	96189.73	148001.21	California	71498.59
43	15505.83	127382.40	35534.27	New York	69759.08
44	22177.84	154806.24	28334.82	California	65200.43
45	1000.33	124153.14	1904.03	New York	64926.18
46	1315.56	115816.31	297114.56	Florida	49490.85
47	0.10	135427.02	0.10	California	42559.83
48	542.15	51743.25	0.10	New York	35673.51
49	0.10	116983.90	45173.16	California	14681.50

In [15]:

```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

Out[15]:

	RD Spend	Administration	Market Spend	Profit	State_Florida	State_New York
0	165349.30	136897.90	471784.20	192261.93	False	True
1	162597.80	151377.69	443898.63	191792.16	False	False
2	153441.61	101145.65	407934.64	191050.49	True	False
3	144372.51	118671.95	383199.72	182902.09	False	True
4	142107.44	91391.87	366168.52	166188.04	True	False
5	131877.00	99814.81	362861.46	156991.22	False	True
6	134615.56	147198.97	127716.92	156122.61	False	False
7	130298.23	145530.16	323876.78	155752.70	True	False

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	RD Spend	Administration	Market Spend	Profit	State_Florida	State_New York
8	120542.62	148719.05	311613.39	152211.87	False	True
9	123334.98	108679.27	304981.72	149760.06	False	False
10	101913.18	110594.21	229161.05	146122.05	True	False
11	100672.06	91790.71	249744.65	144259.50	False	False
12	93863.85	127320.48	249839.54	141585.62	True	False
13	91992.49	135495.17	252665.03	134307.45	False	False
14	119943.34	156547.52	256513.02	132602.75	True	False
15	114523.71	122616.94	261776.33	129917.14	False	True
16	78013.21	121597.65	264346.16	126993.03	False	False
17	94657.26	145077.68	282574.41	125370.47	False	True
18	91749.26	114175.89	294919.67	124267.00	True	False
19	86419.80	153514.21	0.10	122776.96	False	True
20	76253.96	113867.40	298664.57	118474.13	False	False
21	78389.57	153773.53	299737.39	111313.12	False	True
22	73994.66	122782.85	303319.36	110352.35	True	False
23	67532.63	105751.13	304768.83	108734.09	True	False
24	77044.11	99281.44	140574.91	108552.14	False	True
25	64664.81	139553.26	137962.72	107404.44	False	False
26	75328.97	144136.08	134050.17	105733.64	True	False
27	72107.70	127864.65	353183.91	105008.41	False	True
28	66051.62	182645.66	118148.30	103282.48	True	False
29	65605.58	153032.16	107138.48	101004.74	False	True
30	61994.58	115641.38	91131.34	99937.69	True	False
31	61136.48	152702.02	88218.33	97483.66	False	True
32	63408.96	129219.71	46085.35	97427.94	False	False
33	55494.05	103057.59	214634.91	96779.02	True	False
34	46426.17	157694.02	210797.77	96712.90	False	False
35	46014.12	85047.54	205517.74	96479.61	False	True
36	28663.86	127056.31	201126.92	90708.29	True	False
37	44070.05	51283.24	197029.52	89949.24	False	False
38	20229.69	65948.03	185265.20	81229.16	False	True
39	38558.61	82982.19	174999.40	81005.86	False	False
40	28754.43	118546.15	172795.77	78240.01	False	False
41	27893.02	84710.87	164470.81	77798.93	True	False
42	23641.03	96189.73	148001.21	71498.59	False	False

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	RD Spend	Administration	Market Spend	Profit	State_Florida	State_New York
43	15505.83	127382.40	35534.27	69759.08	False	True
44	22177.84	154806.24	28334.82	65200.43	False	False
45	1000.33	124153.14	1904.03	64926.18	False	True
46	1315.56	115816.31	297114.56	49490.85	True	False
47	0.10	135427.02	0.10	42559.83	False	False
48	542.15	51743.25	0.10	35673.51	False	True
49	0.10	116983.90	45173.16	14681.50	False	False

In [25]:

```
x=dataset[["RD Spend","Administration","Market Spend","State_Florida","State_New York"]]
y=dataset[["Profit"]]
print(x)
print(y)
```

```
     RD Spend  Administration  Market Spend  State_Florida  State_New York
0   165349.30       136897.90     471784.20          False            True
1   162597.80       151377.69     443898.63          False           False
2   153441.61       101145.65     407934.64           True           False
3   144372.51       118671.95     383199.72          False            True
4   142107.44        91391.87     366168.52           True           False
5   131877.00        99814.81     362861.46          False            True
6   134615.56       147198.97     127716.92          False           False
7   130298.23       145530.16     323876.78           True           False
8   120542.62       148719.05     311613.39          False            True
9   123334.98       108679.27     304981.72          False           False
10  101913.18       110594.21     229161.05           True           False
11  100672.06        91790.71     249744.65          False           False
12   93863.85       127320.48     249839.54           True           False
13   91992.49       135495.17     252665.03          False           False
14  119943.34       156547.52     256513.02           True           False
15  114523.71       122616.94     261776.33          False            True
16   78013.21       121597.65     264346.16          False           False
17   94657.26       145077.68     282574.41          False            True
18   91749.26       114175.89     294919.67           True           False
19   86419.80       153514.21          0.10          False            True
20   76253.96       113867.40     298664.57          False           False
21   78389.57       153773.53     299737.39          False            True
22   73994.66       122782.85     303319.36           True           False
23   67532.63       105751.13     304768.83           True           False
24   77044.11        99281.44     140574.91          False            True
25   64664.81       139553.26     137962.72          False           False
26   75328.97       144136.08     134050.17           True           False
27   72107.70       127864.65     353183.91          False            True
28   66051.62       182645.66     118148.30           True           False
29   65605.58       153032.16     107138.48          False            True
30   61994.58       115641.38      91131.34           True           False
31   61136.48       152702.02      88218.33          False            True
32   63408.96       129219.71      46085.35          False           False
33   55494.05       103057.59     214634.91           True           False
34   46426.17       157694.02     210797.77          False           False
35   46014.12        85047.54     205517.74          False            True
36   28663.86       127056.31     201126.92           True           False
37   44070.05        51283.24     197029.52          False           False
```

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38	20229.69	65948.03	185265.20	False	True	
39	38558.61	82982.19	174999.40	False	False	
40	28754.43	118546.15	172795.77	False	False	
41	27893.02	84710.87	164470.81	True	False	
42	23641.03	96189.73	148001.21	False	False	
43	15505.83	127382.40	35534.27	False	True	
44	22177.84	154806.24	28334.82	False	False	
45	1000.33	124153.14	1904.03	False	True	
46	1315.56	115816.31	297114.56	True	False	
47	0.10	135427.02	0.10	False	False	
48	542.15	51743.25	0.10	False	True	
49	0.10	116983.90	45173.16	False	False	

	Profit
0	192261.93
1	191792.16
2	191050.49
3	182902.09
4	166188.04
5	156991.22
6	156122.61
7	155752.70
8	152211.87
9	149760.06
10	146122.05
11	144259.50
12	141585.62
13	134307.45
14	132602.75
15	129917.14
16	126993.03
17	125370.47
18	124267.00
19	122776.96
20	118474.13
21	111313.12
22	110352.35
23	108734.09
24	108552.14
25	107404.44
26	105733.64
27	105008.41
28	103282.48
29	101004.74
30	99937.69
31	97483.66
32	97427.94
33	96779.02
34	96712.90
35	96479.61
36	90708.29
37	89949.24
38	81229.16
39	81005.86
40	78240.01
41	77798.93
42	71498.59
43	69759.08
44	65200.43
45	64926.18

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```
46   49490.85
47   42559.83
48   35673.51
49   14681.50
```

In [26]:

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.20,random_state=42)
y_test
```

Out[26]:

	Profit
13	134307.45
39	81005.86
30	99937.69
45	64926.18
17	125370.47
48	35673.51
26	105733.64
25	107404.44
32	97427.94
19	122776.96

In [27]:

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

In [52]:

```
from sklearn.svm import SVR
regressor=SVR(kernel="linear",gamma=100,C=1000)
regressor.fit(x_train, y_train)
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversi
onWarning: A column-vector y was passed when a 1d array was expected. Please change the
shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[52]:

▾ SVR ⓘ ?

▸ Parameters

▸ Fitted attributes

In [53]:

```
y_pred=regressor.predict(x_test)
print(y_pred)
```

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```
[122918.00034996  87218.13006474  92424.14577216  59271.30611261
 131037.190276    54460.77205034 104474.82067104  99526.75359489
  89703.48626418 101212.33341867]
```

In [54]:

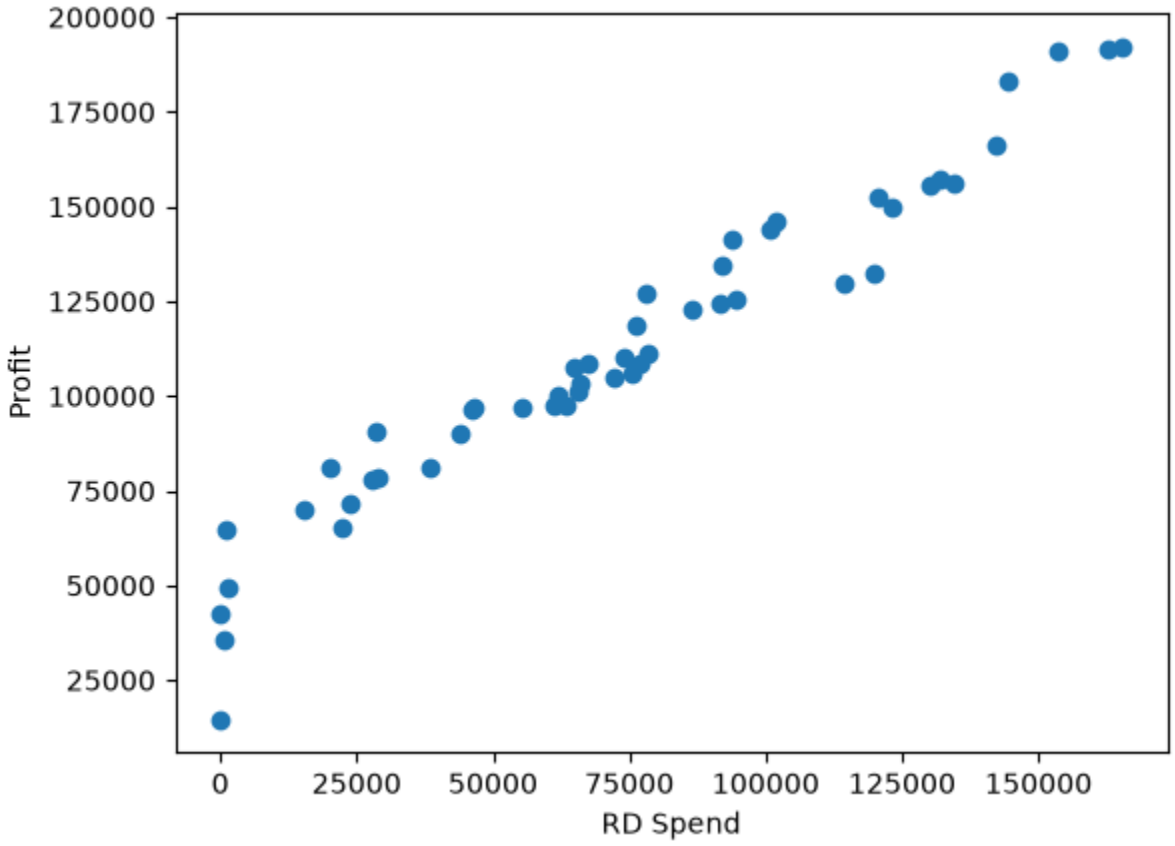
```
from sklearn.metrics import r2_score
result=r2_score(y_pred,y_test)
result
```

Out[54]:

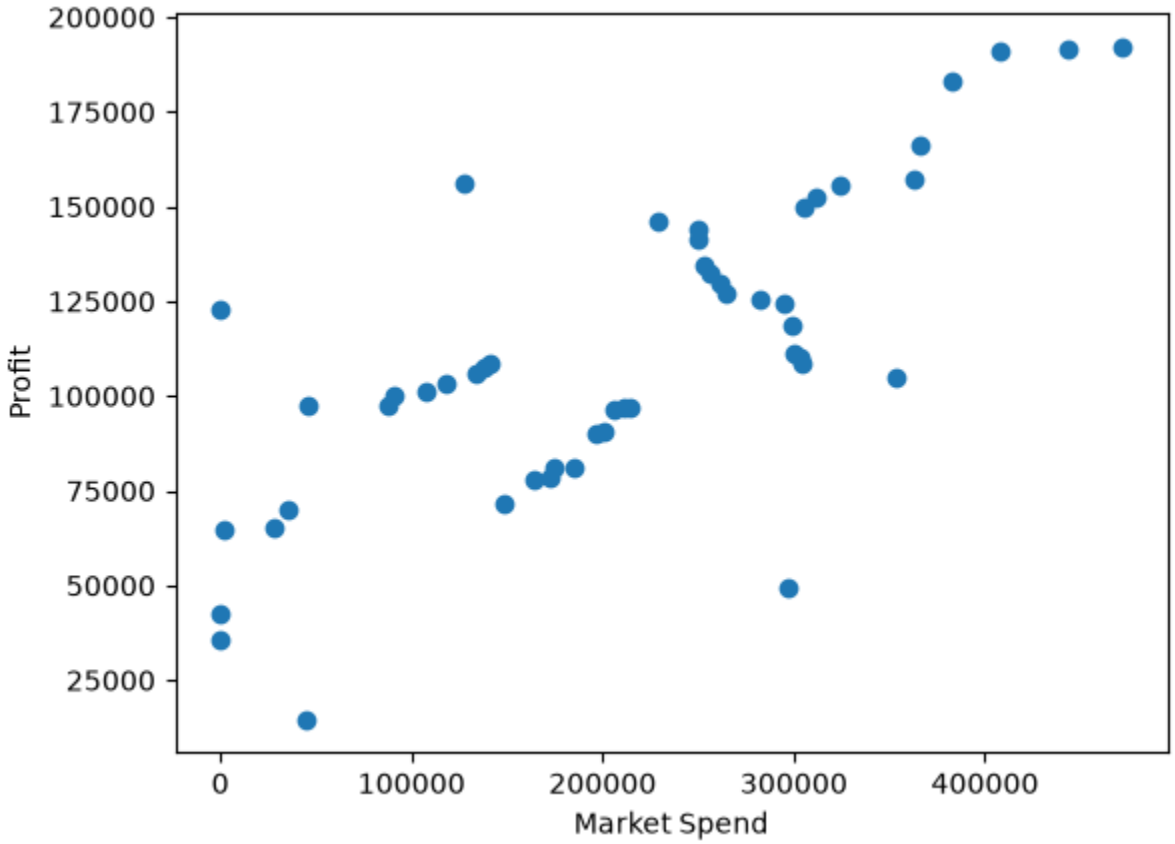
```
0.7650644604778343
```

In [58]:

```
import matplotlib.pyplot as plt
plt.scatter(dataset["RD Spend"],y)
plt.xlabel("RD Spend")
plt.ylabel("Profit")
plt.show()
plt.scatter(dataset["Market Spend"],y)
plt.xlabel("Market Spend")
plt.ylabel("Profit")
plt.show()
```



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